

MATH 231 – Linear Algebra I

The Matrix Unit, Part 1 ■ Linear Systems and the Matrix

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About this document. This opens the unit on *matrices*. It does not begin by defining one. It begins with a question the vector unit left unanswered — given a basis and a vector, how do you find the coordinates? — shows that the question is a system of n linear equations, and then builds the matrix as the notation that system demands.

A common confusion (Section numbers restart). This unit numbers its sections from 1 again. They do *not* continue from the vector unit, whose sections run §1 to §24, and they have nothing to do with the computational unit, which also restarts at 1. So a bare reference such as “§1.3” below means §1.3 *of this unit*. Whenever we point at either of the other two we will name it, as in “the vector unit, §16.5”.

Assumed from the vector unit. $\mathbb{R}^n$ and the bracket notation $\mathbf{v} = [v_1, \dots, v_n]$ (§24.1); addition and scalar multiplication (§7–§8); the dot product $\langle \mathbf{u}, \mathbf{v} \rangle = \sum_i u_i v_i$ (§11.3); linear combinations and span (§16.2–§16.3); coordinates and their uniqueness (§16.4); the coordinate problem for the standard basis (§16.5); linear independence (§18.4); basis and dimension (§16.3, §17); and orthogonality (§11.7).

What you should be able to do afterwards. Say why a basis turns “find the coordinates of $\mathbf{u}$ ” into a system of n linear equations in n unknowns; write that system out from a given basis; say what it means to solve it; recognise each equation as a dot product; build the coefficient matrix $\mathbf{V}$ and the coordinate vector $\mathbf{x}$; state the definition of a matrix–vector product and compute one; write a system as $\mathbf{V}\mathbf{x} = \mathbf{u}$; and read that single equation in both directions — by rows as n dot products, and by columns as one linear combination.

1. The problem a basis leaves behind

1.1. What a basis promises

Work in $\mathbb{R}^n$ and let

$$\mathcal{B} = \{ \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \}$$

be a set of n linearly independent vectors. That is enough to make $\mathcal{B}$ a basis.

From the vector unit. Two facts do the work. A basis for $\mathbb{R}^n$ is a set that spans $\mathbb{R}^n$ and is linearly independent (§16.3), and a basis for $\mathbb{R}^n$ has exactly n vectors, since $\dim \mathbb{R}^n = n$ (§17, §24.1). What is not obvious is that independence alone is enough — that n independent vectors cannot fail to span. The vector unit proved this in $\mathbb{R}^2$ (§18.5) and used it in $\mathbb{R}^n$ (§24.4). We take it as given.

Remark. It is worth being clear that this is a genuine theorem and not a definition. In principle a set could be independent and still miss part of the space; what the result says is that in $\mathbb{R}^n$, once you have n independent vectors, there is no room left for it to miss anything. A proof in general dimension is one of the things the machinery of this unit will eventually supply, which is a fair reason to postpone it rather than assert it twice.

Being a basis, $\mathcal{B}$ spans $\mathbb{R}^n$, and *span* means exactly what it said in the vector unit, §16.3: the set of all linear combinations. So every vector of $\mathbb{R}^n$ is reachable. Written out: for any $\mathbf{u} \in \mathbb{R}^n$ there

are scalars $\alpha_1, \dots, \alpha_n \in \mathbb{R}$ with

$$\mathbf{u} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_n \mathbf{v}_n = \sum_{i=1}^n \alpha_i \mathbf{v}_i$$

and, because coordinates against a basis are unique (§16.4), there is exactly one such list of scalars.

That is a promise about existence, and it is the kind of promise that is easy to undervalue. It says: pick any point of $\mathbb{R}^n$ you like — any point at all, however far out, in however many dimensions — and it can be built out of these n vectors, in one and only one way.

What the promise does *not* do is tell you the scalars.

1.2. The reverse problem

So we ask for them. This is the reverse problem of the vector unit, §16.5, now posed in $\mathbb{R}^n$ and against an arbitrary basis rather than the standard one:

Given a basis $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ and a vector $\mathbf{u} \in \mathbb{R}^n$, find the scalars $\alpha_1, \dots, \alpha_n$ with $\mathbf{u} = \alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n$.

The $\mathbf{v}_i$ are known. The vector $\mathbf{u}$ is known. The α_i are the unknowns, and §1.1 has already told us two things about them before we begin: they exist, and there is only one list of them. We are not searching hopefully. We are computing something we know is there and know is unambiguous — which is precisely the situation in which it is worth developing a method.

From the vector unit. When $\mathcal{B}$ was the standard basis this question was free. Because $\mathbf{e}_i$ has a 1 in slot i and zeros elsewhere, the coordinates of $\mathbf{u}$ are just its components, $\alpha_i = u_i$, read off with no work at all (§16.5 in $\mathbb{R}^2$, §24.1 in $\mathbb{R}^n$). The vector unit was careful to say why: the standard basis is orthonormal, and the zeros that make the reading free are the statement $\langle \mathbf{e}_i, \mathbf{e}_j \rangle = 0$. Change the basis and the zeros go, and with them the free answer.

1.3. The system, written out

To turn the vector equation into something we can compute with, expand it componentwise. Two vectors in $\mathbb{R}^n$ are equal exactly when all n of their components agree (§9), so one vector equation is n scalar equations.

First, notation for the components of the basis vectors. Each $\mathbf{v}_i$ is a vector in $\mathbb{R}^n$, so it has n components, and we need a symbol carrying both *which vector* and *which slot*:

$$\mathbf{v}_i = [v_{1i}, v_{2i}, \dots, v_{ni}], \quad \text{so} \quad v_{ki} = \text{the } k\text{-th component of } \mathbf{v}_i.$$

A common confusion (Which index is which). Read v_{ki} as “slot k , vector i ”. The *first* index says which component and the *second* says which basis vector — the opposite of the order in which you would say it out loud. This is deliberate, and §2.2 is where it pays: written this way, v_{ki} will turn out to sit in row k and column i of the object we are about to build, so the two indices are the two directions of a grid, in that order. Choose the other convention and you spend the rest of the unit transposing in your head.

Now expand. Scaling $\mathbf{v}_i$ by α_i multiplies every component (§8), and adding vectors adds them slot by slot (§7), so slot k of the right-hand side collects one term from each basis vector:

$$\text{slot } k \text{ of } \sum_{i=1}^n \alpha_i \mathbf{v}_i = \alpha_1 v_{k1} + \alpha_2 v_{k2} + \dots + \alpha_n v_{kn}.$$

Setting that equal to u_k for each $k = 1, \dots, n$ gives n equations:

$$\begin{aligned}\alpha_1 v_{11} + \alpha_2 v_{12} + \cdots + \alpha_n v_{1n} &= u_1, \\ \alpha_1 v_{21} + \alpha_2 v_{22} + \cdots + \alpha_n v_{2n} &= u_2, \\ &\vdots \\ \alpha_1 v_{n1} + \alpha_2 v_{n2} + \cdots + \alpha_n v_{nn} &= u_n.\end{aligned}$$

Definition 1.1 (Linear system). A **linear system** is a finite collection of linear equations in the same unknowns, to be satisfied *simultaneously*. The display above is a system of n equations in the n unknowns $\alpha_1, \dots, \alpha_n$.

Read the display in both directions once, because both readings are used constantly.

- **Across a row.** Row k is one equation. It involves every unknown, and the coefficients it uses are the k -th components of all n basis vectors.
- **Down a column.** Column i belongs to one basis vector. Reading $v_{1i}, v_{2i}, \dots, v_{ni}$ downward reassembles $\mathbf{v}_i$ itself.

That the same array of numbers organises into rows that are equations and columns that are vectors is not a coincidence, and it is the whole reason §2 works.

1.4. What it means to solve it

Definition 1.2 (Solving a linear system). To **solve** a linear system is to find values of the unknowns that satisfy *every* equation at once. A list of values satisfying all of them is a **solution**.

The word *simultaneously* is the entire difficulty. Any one of those n equations is easy on its own — one equation in n unknowns has endlessly many solutions, and you can satisfy it by picking $n - 1$ of the α 's at random and solving for the last. What is hard is finding the one choice that satisfies all n at the same time. Each equation is a constraint, and the solution is what survives all of them together.

Remark. For the system of §1.3 we already know the answer set has exactly one element, because $\mathcal{B}$ is a basis and coordinates are unique. That is a strong piece of information and it will not always be available. Systems in general can have one solution, no solution, or infinitely many, and deciding which is the case is a question in its own right. Here the basis hypothesis has settled it in advance, which makes this the cleanest possible first example of a linear system.

1.5. A worked case in $\mathbb{R}^2$

Take $n = 2$ and the basis

$$\mathbf{v}_1 = [2, 1], \quad \mathbf{v}_2 = [1, 3],$$

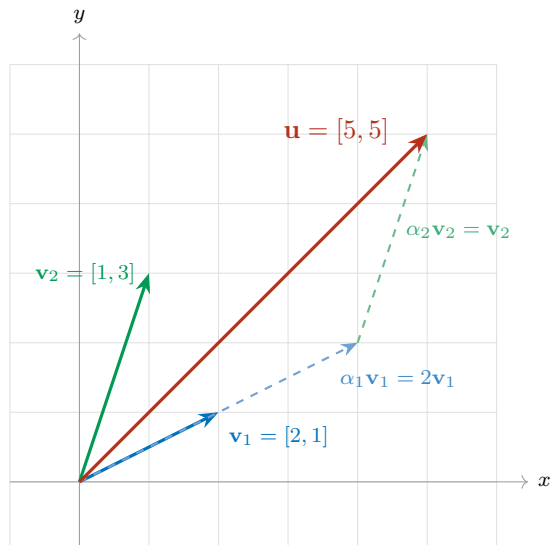
which is independent: if $\alpha_1[2, 1] + \alpha_2[1, 3] = \mathbf{0}$ then $2\alpha_1 + \alpha_2 = 0$ and $\alpha_1 + 3\alpha_2 = 0$; the first gives $\alpha_2 = -2\alpha_1$, and substituting into the second gives $\alpha_1 - 6\alpha_1 = -5\alpha_1 = 0$, so $\alpha_1 = \alpha_2 = 0$. Two independent vectors in $\mathbb{R}^2$, so a basis.

Let $\mathbf{u} = [5, 5]$ and ask for its coordinates. In the notation of §1.3, $v_{11} = 2$, $v_{21} = 1$, $v_{12} = 1$, $v_{22} = 3$, and the system is

$$\begin{aligned}2\alpha_1 + 1\alpha_2 &= 5, \\ 1\alpha_1 + 3\alpha_2 &= 5.\end{aligned}$$

Solve it by substitution: the first equation gives $\alpha_2 = 5 - 2\alpha_1$, and putting that into the second gives $\alpha_1 + 15 - 6\alpha_1 = 5$, so $-5\alpha_1 = -10$ and $\alpha_1 = 2$; then $\alpha_2 = 5 - 4 = 1$. Checking against the original vector equation,

$$2\mathbf{v}_1 + 1\mathbf{v}_2 = 2[2, 1] + [1, 3] = [4, 2] + [1, 3] = [5, 5] = \mathbf{u}. \quad \checkmark$$



The picture is the same walk as in the vector unit, §16.5 — travel α_1 steps along the first basis vector, then α_2 along the second, and arrive at the head of $\mathbf{u}$ — but the steps are no longer along the axes, and that is exactly what makes the arithmetic non-trivial.

Remark (What changed, and it is only one thing). Against the standard basis the same system would have read $1\alpha_1 + 0\alpha_2 = u_1$ and $0\alpha_1 + 1\alpha_2 = u_2$, and the zeros would have collapsed it on sight. Here there are no zeros: both unknowns appear in both equations, so neither can be found without the other, and the equations have to be handled together. The vector unit predicted this at the end of §16.5. Substitution coped with two equations; it is not a method that survives twenty.

1.6. The same thing one dimension up

One more, in $\mathbb{R}^3$, to have a case with enough room to show the shape of things. Take

$$\mathbf{v}_1 = [1, 0, 1], \quad \mathbf{v}_2 = [2, 1, 0], \quad \mathbf{v}_3 = [0, 1, 1].$$

These are independent, and checking it is itself a small system: setting $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \alpha_3\mathbf{v}_3 = \mathbf{0}$ and reading the three slots gives $\alpha_1 + 2\alpha_2 = 0$, $\alpha_2 + \alpha_3 = 0$ and $\alpha_1 + \alpha_3 = 0$. The last two give $\alpha_1 = \alpha_2 = -\alpha_3$, and substituting into the first gives $-3\alpha_3 = 0$, so all three are zero. Three independent vectors in $\mathbb{R}^3$, so a basis.

Now suppose the coordinates of some $\mathbf{u}$ turn out to be $\alpha_1 = 1$, $\alpha_2 = 2$, $\alpha_3 = 3$. Then

$$\mathbf{u} = 1[1, 0, 1] + 2[2, 1, 0] + 3[0, 1, 1] = [1, 0, 1] + [4, 2, 0] + [0, 3, 3] = [5, 5, 4].$$

Read off the components with the convention of §1.3 — v_{ki} is slot k of $\mathbf{v}_i$, so $v_{12} = 2$ is the first slot of the second vector — and the system for the unknown coordinates of $\mathbf{u} = [5, 5, 4]$ is

$$\begin{aligned} 1\alpha_1 + 2\alpha_2 + 0\alpha_3 &= 5, \\ 0\alpha_1 + 1\alpha_2 + 1\alpha_3 &= 5, \\ 1\alpha_1 + 0\alpha_2 + 1\alpha_3 &= 4, \end{aligned}$$

whose solution is $\alpha_1 = 1, \alpha_2 = 2, \alpha_3 = 3$, as constructed. This is the example §2 will carry through. Notice already what §1.3 promised: the coefficients going *down* the first column are 1, 0, 1, which is $\mathbf{v}_1$; down the second, 2, 1, 0, which is $\mathbf{v}_2$; down the third, 0, 1, 1, which is $\mathbf{v}_3$.

1.7. Where else these come from

The coordinate problem is how linear systems enter *this* course, and it is a good entrance because everything about it is already understood: we know a solution exists, we know it is unique, and we know what it means. It is not the only entrance. Systems of linear equations arise whenever several linear constraints must hold at once — balancing a chemical reaction, fitting a line through data, computing currents in a circuit, allocating a fixed budget across competing demands, enforcing conservation at every node of a network.

What all of these share is not their subject matter but their *shape*: a list of unknowns, and a list of linear equations relating them. That shape is what the next section names, and naming it is what lets one method serve all of them.

2. Compressing the notation

The system of §1.3 is correct and unwieldy. Writing it out costs n lines, each with n terms, so a system in twenty unknowns fills a page before any work is done — and none of that page is doing anything a symbol could not.

Before asking *how* to solve such a system, we spend this section building notation for it. Four steps, each one small.

2.1. Step 1: every equation is a dot product

Look at a single equation of the system — row k :

$$\alpha_1 v_{k1} + \alpha_2 v_{k2} + \cdots + \alpha_n v_{kn} = u_k.$$

The left-hand side is a sum of products of paired-up numbers, which is a structure we have a name for. Set

$$\mathbf{r}_k = [v_{k1}, v_{k2}, \dots, v_{kn}], \quad \mathbf{x} = [\alpha_1, \alpha_2, \dots, \alpha_n],$$

and the left-hand side is precisely $\langle \mathbf{r}_k, \mathbf{x} \rangle$, the dot product of the vector unit, §11.3. So the system is

$$\langle \mathbf{r}_1, \mathbf{x} \rangle = u_1, \quad \langle \mathbf{r}_2, \mathbf{x} \rangle = u_2, \quad \dots, \quad \langle \mathbf{r}_n, \mathbf{x} \rangle = u_n.$$

Nothing has been computed and nothing has been assumed. We have only noticed that an operation we already have describes each line.

Example 2.1. For the $\mathbb{R}^3$ system of §1.6, the three vectors $\mathbf{r}_k$ are $[1, 2, 0]$, $[0, 1, 1]$ and $[1, 0, 1]$, and with $\mathbf{x} = [1, 2, 3]$,

$$\langle [1, 2, 0], [1, 2, 3] \rangle = 1 + 4 + 0 = 5,$$

$$\langle [0, 1, 1], [1, 2, 3] \rangle = 0 + 2 + 3 = 5,$$

$$\langle [1, 0, 1], [1, 2, 3] \rangle = 1 + 0 + 3 = 4,$$

which are the three right-hand sides 5, 5, 4. ■

Definition 2.1 (Coordinate vector). The vector $\mathbf{x} = [\alpha_1, \dots, \alpha_n]$ holding the unknown coordinates is the **coordinate vector** of $\mathbf{u}$ with respect to $\mathcal{B}$.

Remark. Two different vectors of $\mathbb{R}^n$ are now in play and it is worth keeping them apart. The vector $\mathbf{u}$ is a point of the space. The vector $\mathbf{x}$ is the recipe for building it out of $\mathcal{B}$. They are equal only when $\mathcal{B}$ is the standard basis — which is the content of the vector unit’s remark at §16.4, that the bracket notation was silently reporting coordinates in one particular basis all along.

2.2. Step 2: the matrix

The n vectors $\mathbf{r}_1, \dots, \mathbf{r}_n$ carry all the coefficients of the system, and $\mathbf{x}$ carries all the unknowns. Stack the $\mathbf{r}_k$ in order, one per line, keeping their entries aligned in columns.

Definition 2.2 (Matrix). A **matrix** is a rectangular array of numbers arranged in rows and columns. The matrix of the system of §1.3 is

$$\mathbf{V} = \begin{bmatrix} v_{11} & v_{12} & \cdots & v_{1n} \\ v_{21} & v_{22} & \cdots & v_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ v_{n1} & v_{n2} & \cdots & v_{nn} \end{bmatrix},$$

with n rows and n columns. The number v_{ki} in row k and column i is the (k, i) **entry** of $\mathbf{V}$.

This is the second object of the course, after the vector, and the convention of §1.3 is what makes it painless to write down: the two indices of v_{ki} are already the two directions of the grid, row first and column second. Nothing had to be rearranged.

Now the two readings of §1.3 become geometric facts about the array.

- **Row k of $\mathbf{V}$ is $\mathbf{r}_k$** , the coefficients of the k -th equation. Rows are equations.
- **Column i of $\mathbf{V}$ is $\mathbf{v}_i$** , the i -th basis vector, written downward. Columns are the basis.

So $\mathbf{V}$ is the basis $\mathcal{B}$ with its vectors stood up side by side:

$$\mathbf{V} = \left[\mathbf{v}_1 \mid \mathbf{v}_2 \mid \cdots \mid \mathbf{v}_n \right],$$

which is worth remembering as the picture of what the matrix *is*, before it is anything else.

Example 2.2. The $\mathbb{R}^3$ system of §1.6 has

$$\mathbf{V} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

Its rows $[1, 2, 0]$, $[0, 1, 1]$, $[1, 0, 1]$ are the three equations. Its columns, read downward, are $[1, 0, 1]$, $[2, 1, 0]$, $[0, 1, 1]$ — exactly $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. ■

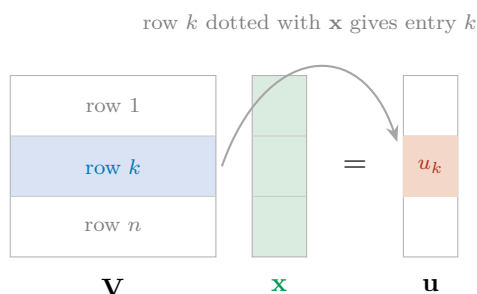
2.3. Step 3: multiplying a matrix by a vector

We have a matrix holding the coefficients and a vector holding the unknowns, and §2.1 said what to do with them: dot each row against $\mathbf{x}$. Doing that for every row produces n numbers, which is a vector. So the operation takes a matrix and a vector and returns a vector, and it deserves a name and a symbol.

Definition 2.3 (Matrix–vector product). Let $\mathbf{V}$ be an $n \times n$ matrix with rows $\mathbf{r}_1, \dots, \mathbf{r}_n$, and let $\mathbf{x} \in \mathbb{R}^n$. The **matrix–vector product** $\mathbf{V}\mathbf{x}$ is the vector of $\mathbb{R}^n$ whose k -th component is the dot product of row k with $\mathbf{x}$:

$$\mathbf{V}\mathbf{x} = \begin{bmatrix} \langle \mathbf{r}_1, \mathbf{x} \rangle \\ \langle \mathbf{r}_2, \mathbf{x} \rangle \\ \vdots \\ \langle \mathbf{r}_n, \mathbf{x} \rangle \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^n v_{1i} \alpha_i \\ \sum_{i=1}^n v_{2i} \alpha_i \\ \vdots \\ \sum_{i=1}^n v_{ni} \alpha_i \end{bmatrix}.$$

In words, and this is the sentence to memorise: *to multiply a matrix by a vector, take the dot product of each row of the matrix with that vector*. The row supplies one component of the answer, and there are as many components as rows.



A common confusion (Rows, columns, and the bracket notation). The vector unit wrote vectors horizontally, $\mathbf{v} = [v_1, \dots, v_n]$, and the display above writes $\mathbf{x}$ and $\mathbf{V}\mathbf{x}$ vertically. Nothing has changed about the vectors. The bracket notation never claimed a shape — it listed n numbers in order, which is all a vector in $\mathbb{R}^n$ is — and $[1, 2, 3]$ and its column spelling denote the same element of $\mathbb{R}^3$.

The reason for switching is that $\mathbf{V}\mathbf{x}$ dots $\mathbf{x}$ against each *row*, so drawing $\mathbf{x}$ as a column puts each of its entries directly beneath the column of $\mathbf{V}$ it multiplies, and the arithmetic can be read off the page. Horizontal brackets stay in use for prose, where they take less room. When the distinction between a row and a column starts to carry mathematical weight, we will say so; here it is typography.

Example 2.3. With $\mathbf{V}$ from §2.2 and $\mathbf{x} = [1, 2, 3]$:

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1(1) + 2(2) + 0(3) \\ 0(1) + 1(2) + 1(3) \\ 1(1) + 0(2) + 1(3) \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \\ 4 \end{bmatrix}.$$

Three dot products, three components. ■

Remark (A shape rule, for later). Row k has n entries and $\mathbf{x}$ has n entries, and a dot product is defined only when both vectors have the same dimension (the vector unit, §11.3). So $\mathbf{V}\mathbf{x}$ makes sense only when the number of *columns* of $\mathbf{V}$ matches the number of entries of $\mathbf{x}$. Every matrix in this part is $n \times n$ and the condition is automatic, but it is the rule that will govern all matrix multiplication later, and it comes from nowhere more exotic than the dot product needing two vectors of equal length.

2.4. Step 4: $\mathbf{V}\mathbf{x} = \mathbf{u}$

Now put the three pieces together. By §2.3, the k -th component of $\mathbf{V}\mathbf{x}$ is $\langle \mathbf{r}_k, \mathbf{x} \rangle$. By §2.1, the k -th equation of the system says that this equals u_k . Two vectors are equal exactly when all their components agree. So the entire system — all n equations, all n unknowns — is the single statement

$$\boxed{\mathbf{V}\mathbf{x} = \mathbf{u}}$$

That is the compression. The $\mathbb{R}^3$ example of §1.6, which took three lines and nine coefficients to state, is

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \\ 4 \end{bmatrix},$$

and in $\mathbb{R}^{20}$ it would still be $\mathbf{V}\mathbf{x} = \mathbf{u}$.

Symbol	What it is	Known?
$\mathbf{V}$	the $n \times n$ matrix whose columns are the basis vectors	known
$\mathbf{x}$	the coordinate vector $[\alpha_1, \dots, \alpha_n]$	unknown
$\mathbf{u}$	the vector whose coordinates we want	known

Compare the shape of $\mathbf{V}\mathbf{x} = \mathbf{u}$ with the scalar equation $ax = b$, whose solution is $x = b/a$ whenever $a \neq 0$. The resemblance is not accidental and it is not yet a method: we have no division of vectors, and no meaning for $\mathbf{V}^{-1}$. But the resemblance is the reason the notation is worth having. It makes a question of n equations look like a question about one object, and questions about one object are the kind we know how to ask.

Remark (What was actually gained). Three things, and only the first is about saving ink.

Brevity. One symbol for n^2 coefficients.

Uniformity. The coordinate problem, a circuit, a least-squares fit and a network flow all reduce to $\mathbf{V}\mathbf{x} = \mathbf{u}$ with different $\mathbf{V}$. A method for the equation is a method for all of them at once.

A new object. $\mathbf{V}$ was introduced as bookkeeping, but it will not stay bookkeeping. Matrices add, multiply, and have their own arithmetic; a matrix will turn out to *act* on vectors rather than merely tabulate numbers, and $\mathbf{V}\mathbf{x}$ will be read as a function applied to $\mathbf{x}$. That reading is where most of the rest of this course lives.

2.5. The same equation, read by columns

The definition in §2.3 built $\mathbf{V}\mathbf{x}$ out of rows. There is a second way to read the same product, and it closes the loop back to §1.1.

Take the $\mathbb{R}^3$ computation again, and instead of collecting the arithmetic into three dot products, group it by which column of $\mathbf{V}$ each term came from:

$$\begin{bmatrix} 1(1) + 2(2) + 0(3) \\ 0(1) + 1(2) + 1(3) \\ 1(1) + 0(2) + 1(3) \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

The left side is grouped by row, the right side by column, and both are the same nine products added in a different order. But the right-hand side is $\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \alpha_3 \mathbf{v}_3$ — a linear combination of the basis vectors, with the entries of $\mathbf{x}$ as the weights.

Theorem 2.1 (The two readings of a matrix–vector product). *Let $\mathbf{V}$ have columns $\mathbf{v}_1, \dots, \mathbf{v}_n$ and let $\mathbf{x} = [\alpha_1, \dots, \alpha_n]$. Then*

$$\mathbf{V}\mathbf{x} = \underbrace{[\langle \mathbf{r}_1, \mathbf{x} \rangle, \dots, \langle \mathbf{r}_n, \mathbf{x} \rangle]}_{\text{by rows}} = \underbrace{\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_n \mathbf{v}_n}_{\text{by columns}}.$$

Proof. Compare slot k of each side. On the left it is $\langle \mathbf{r}_k, \mathbf{x} \rangle = \sum_{i=1}^n v_{ki} \alpha_i$, by §2.3. On the right, slot k of $\alpha_i \mathbf{v}_i$ is $\alpha_i v_{ki}$, since scaling acts componentwise, and adding the n vectors adds their k -th slots, giving $\sum_{i=1}^n \alpha_i v_{ki}$. The two sums have the same terms. $\square$

So $\mathbf{V}\mathbf{x} = \mathbf{u}$ says, in the column reading,

$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_n \mathbf{v}_n = \mathbf{u},$$

which is the boxed equation of §1.1 — the one we started from. We have gone around a full circle and arrived back where we began, in new notation.

Remark (Why the circle was worth walking). Nothing was proved on the trip; the two ends are the same statement. What changed is that the statement now has a *form*, and the form is what can be worked on. “Find weights that combine these vectors into $\mathbf{u}$ ” is a description. “Solve $\mathbf{V}\mathbf{x} = \mathbf{u}$ ” is a problem with an object in it.

Keep both readings. The row reading is how you *compute* a matrix–vector product, and it is what an algorithm implements. The column reading is how you *understand* one: $\mathbf{V}\mathbf{x}$ is a linear combination of the columns of $\mathbf{V}$. Most of the structural theorems ahead — when a system has a solution, how many, what the columns of a matrix span — are consequences of the column reading.

A common confusion ($\mathbf{V}$ is not a list of the basis vectors). Writing $\mathbf{V}$ as its columns is a picture, not an identification. A basis is a *set* — $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$, unordered, with no repeats. A matrix is an *array*, in which order matters and entries may repeat. Swapping two columns of $\mathbf{V}$ gives a different matrix but the same basis, with the coordinates permuted to match. So $\mathbf{V}$ records the basis *together with* a choice of ordering for it, which the set $\mathcal{B}$ alone does not supply.

2.6. Practice

P1. Let $\mathbf{v}_1 = [3, 1]$ and $\mathbf{v}_2 = [1, 2]$, and let $\mathbf{u} = [7, 4]$. Write the system for the coordinates of $\mathbf{u}$, solve it, and check your answer in the original vector equation.

P2. Write the matrix $\mathbf{V}$ for the basis in P1, and verify that $\mathbf{V}\mathbf{x} = \mathbf{u}$ for the $\mathbf{x}$ you found.

P3. Compute $\begin{bmatrix} 2 & -1 & 0 \\ 1 & 1 & 3 \\ 0 & 4 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ twice: once by the row reading, once by the column reading.

The answers must agree.

P4. Let $\mathcal{B} = \{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ be the standard basis of $\mathbb{R}^n$. Write down $\mathbf{V}$, and use §2.5 to say in one sentence why the system $\mathbf{V}\mathbf{x} = \mathbf{u}$ has solution $\mathbf{x} = \mathbf{u}$.

P5. Suppose $\mathbf{v}_1 = [1, 2]$ and $\mathbf{v}_2 = [2, 4]$. Show that the system for the coordinates of $\mathbf{u} = [3, 1]$ has no solution, and say which hypothesis of §1.1 fails.

Answer key

P1. The system is $3\alpha_1 + \alpha_2 = 7$ and $\alpha_1 + 2\alpha_2 = 4$. From the first, $\alpha_2 = 7 - 3\alpha_1$; substituting, $\alpha_1 + 14 - 6\alpha_1 = 4$, so $-5\alpha_1 = -10$ and $\alpha_1 = 2$, then $\alpha_2 = 1$. Check: $2[3, 1] + 1[1, 2] = [6, 2] + [1, 2] = [7, 4]$. ✓

P2. The columns of $\mathbf{V}$ are the basis vectors, so $\mathbf{V} = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$ and $\mathbf{x} = [2, 1]$. Row 1: $3(2) + 1(1) = 7$. Row 2: $1(2) + 2(1) = 4$. So $\mathbf{V}\mathbf{x} = [7, 4] = \mathbf{u}$. ✓

P3. By rows: $2(1) - 1(2) + 0(-1) = 0$; $1(1) + 1(2) + 3(-1) = 0$; $0(1) + 4(2) - 2(-1) = 10$. So the product is $[0, 0, 10]$. By columns:

$$1 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} -1 \\ 1 \\ 4 \end{bmatrix} + (-1) \begin{bmatrix} 0 \\ 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} -2 \\ 2 \\ 8 \end{bmatrix} + \begin{bmatrix} 0 \\ -3 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix}. \quad \checkmark$$

P4. The columns are $\mathbf{e}_1, \dots, \mathbf{e}_n$, so $\mathbf{V}$ has 1s down the diagonal and 0s elsewhere. By the column reading, $\mathbf{V}\mathbf{x} = \alpha_1\mathbf{e}_1 + \dots + \alpha_n\mathbf{e}_n = [\alpha_1, \dots, \alpha_n] = \mathbf{x}$, so the equation reads $\mathbf{x} = \mathbf{u}$ directly. This is the vector unit's observation at §24.1 that coordinates against the standard basis are the components themselves, now visible as a property of one particular matrix.

P5. The system is $\alpha_1 + 2\alpha_2 = 3$ and $2\alpha_1 + 4\alpha_2 = 1$. The second equation is twice the left-hand side of the first, so it demands $2(3) = 1$, which is false; no pair (α_1, α_2) satisfies both. The hypothesis that fails is *independence*: $\mathbf{v}_2 = 2\mathbf{v}_1$, so $\mathcal{B}$ is linearly dependent (the vector unit, §18.4), hence not a basis, and its span is a line rather than $\mathbb{R}^2$ — a line that does not contain $[3, 1]$. With the basis hypothesis gone, so is the guarantee of §1.1 that a solution exists at all.

Looking ahead. We can now state the problem in one line but not yet solve it, and substitution — fine for the 2×2 case of §1.5 — is not a method that scales. What is needed is a procedure that works on $\mathbf{V}$ directly, applies to any size, and terminates with the answer or with a definite report that there is none. That procedure is *Gaussian elimination*, and it is next. With it come the questions that P5 has already raised: when does a system have a solution, when is it unique, and what does the matrix look like in each case.